Helwan International Technological University

Technological Faculty in Cairo

Program: Industrial control Technology

Level: One

Subject: Mathematics 1

Sheet 2

1. Solve the following equations in one variable:

a. $2x + 1 = -9$ **b.** $-2(3x - 1) + x = 14 - x$ **c.** $\frac{3}{x} - \frac{1}{3} = \frac{1}{6}$

2. Solve the following quadratic equations:

a. $x^2 - 5x - 6 = 0$ **b.** $x^2 - 4x - 21 = 0$ **c.** $3(x - 4)^2 = 12$

3. Solve the quadratic equations using the quadratic formula:

a. $9x^2 + 3x - 2 = 0$ **b.** $x^2 + x = 4$

4. Solve by factoring:

a. $12x^4 = 3x^2$ **b.** $\sqrt{x + 3} = 3x - 1$ **c.** $2x^5 - 14x^3 = 0$

Choose the correct answer for the following question:

5	**If $x^2 = 25$, then x= …..**			
	5	-5	±5	0
6	**The quadratic Formula:**			
	$x = \frac{-b \pm \sqrt{b^2 - ac}}{2a}$	$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$	$x = \frac{-a \pm \sqrt{b^2 - 4ac}}{2a}$	$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{a}$
7	**Which of the following Radical Equation:**			
	3x +1 = 10	$\sqrt{2 + x} = 4$	$x^2 + x = 2$	None
8	**If x(x-4) = 2 , then x= …..**			
	x= 2 and x= 6	x= -2 and x= 2	x= 2	x= 6

9	**Linear equations in one variable may take the form:**			
	$ax^3+bx^2+cx=0$	$ax^2+bx+c=0$	$ax^2+b=0$	$ax+b=0$
10	**The following equation $x^3+x^2-9x-9=0$ is called ……. equation**			
	radical	polynomial	quadratic	None
11	**Th …….. consists of all values that make the equation true**			
	solution set	real numbers	degree	Leading term
12	**An …… equation is true for all values of the variable.**			
	conditional	radical	polynomial	identity